

COSC 3337 — Homework 1

YOUR NAME HERE

Part 1 — Relational algebra

RA1

$$\Pi_{\text{title}}(\sigma_{\text{duration.sec} < 999}(\text{track}))$$

Resulting tuples:

- ...

RA2

Your explanation here: how many tuples, how many rows in `artist`, and why they differ.

RA3

Resulting tuples:

- ...

RA4

Resulting tuples:

- ...

RA5

Resulting tuples:

- ...

RA6

- (a) Number of tuples, and how you got it:
- (b) What is wrong with the expression:
- (c) A correct expression:

Resulting tuples:

- ...

RA7

- (a) Equivalent? Justification:
- (b) Equivalent? Justification, and the general rule: